FranklinWH-Automation Setup Checklist

For the configuration review and first-time setup · version 4.7.0-rc.6

Use this list to gather everything the setup asks for before you start. Each item says whether it is required, what it is for, and where to find it. The same explanations appear in the (i) pop-ups on the review screens.

Filling it in. Type into the boxes in any PDF reader (Adobe Reader, Preview, a browser) and save a copy, or print it and write by hand. Everything is optional to write down; the point is to have it at hand.

Passwords and keys. Items marked with an "I have this" box are credentials. They stay in the .env file on your machine and the review never displays them. It is safer to tick the box than to write the value here. If you do write them, keep this file private.

Have this ready before you commit. The setup refuses to save until every required item is entered, because the engine cannot run safely on a guess: your FranklinWH login and gateway ID; the site latitude and longitude; the number of batteries and their total capacity (detected when the aGate is reachable); how your exports are credited; for each solar array its name, type, AC size and connection details; and your rate plan with every month of the year in a season, each season's prices, and the time windows when each price applies. A feature you turn on (weather, PVOutput, SolarEdge portal, solar forecasting) needs its own key or details too.

Checking against the review. After you gather the information, open the dashboard, go to Settings and choose *Configuration review*. Step through the screens and tick *Matches review* for each item that shows what you expected. Nothing is saved until you press Commit on the last step (*Preview* never saves).

REQUIRED	The system cannot be set up without it.
RECOMMENDED	Works without it, but strongly advised.
OPTIONAL	Adds a feature; skip if you do not use it.
IF APPLIES	Required only when the condition shown under the item applies.

Before you start

The automation runs in Docker on a computer that stays on around the clock (a NAS, mini PC or Raspberry Pi-class machine) on the same home network as your FranklinWH aGate.

Item	Need	Your information	Matches review
Machine that will run the automation	REQUIRED		☐
What it's for: Runs the container 24/7. It needs Docker with Compose, and it must be on the same local network as the aGate. **Where to find it:** Any always-on computer you control. Note its name or IP so you can open the dashboard at http://<that address>:8100 afterwards.			
Time zone	REQUIRED		☐
What it's for: Peak windows, schedules and daily rollups all run on local time. **Where to find it:** Your IANA time zone name, for example America/Los_Angeles or America/Chicago. Set in the Docker environment, not in the review.			

1. Connection

Your FranklinWH account, and how the automation talks to the aGate.

Item	Need	Your information	Matches review
FranklinWH account email	REQUIRED		☐
What it's for: The automation signs in to the FranklinWH cloud as a backup when the local connection is unavailable. A new sign-in ends the previous one on the same account, so if you also run Home Assistant's FranklinWH integration, give this tool its own FranklinWH account (or app user) or the two keep signing each other out. **Where to find it:** The email you use to log in to the FranklinWH app, or the separate user you added for the automation.			

FranklinWH account password REQUIRED

☐ I have this (safer: tick this instead of writing it down)

What it's for: Used with the email above. It stays in the .env file on your machine; the review only shows whether it is set.
Where to find it: Your FranklinWH app password.

aGate gateway ID REQUIRED

What it's for: Identifies which aGate on your account to control. A 20-character code such as 10060005A02X24470437.
Where to find it: FranklinWH app: Settings, then System Info, then Gateway ID (menu names can vary by app version). It is also the aGate serial on the unit label.

aGate IP address RECOMMENDED

What it's for: Where to reach the aGate on your home network. Used by the local connection and by Modbus.
Where to find it: Your router's list of connected devices / DHCP clients; look for the FranklinWH or aGate entry. Give it a DHCP reservation in the router so the address never changes. A wired (Ethernet) aGate is more reliable than Wi-Fi. If you leave it blank, the review's Test button searches your network for the aGate.

Local connection turned on? RECOMMENDED ☐ Yes ☐ No ☐ Not sure

What it's for: The automation reads the aGate and changes modes directly over your home network. It keeps working when the FranklinWH cloud is down; the cloud is only a backup. On by default. Strongly recommended.
Where to find it: Nothing to set up in the FranklinWH app. In the review, press Test on the Connection step to check it reaches your aGate.

Modbus turned on? OPTIONAL ☐ Yes ☐ No ☐ Not sure

What it's for: A second local channel (port 502). It double-checks every reading and mode change from the local connection and fills in grid voltage, frequency and temperatures. Optional: the local connection works without it.
Where to find it: Modbus is not a setting most owners can reach in the FranklinWH app. If you want it, ask your installer or FranklinWH support whether Modbus TCP can be enabled on your aGate. Leave it off otherwise.

2. Battery

Battery count and capacity are detected automatically. Write them down so you can confirm the detection is right.

Item	Need	Your information	Matches review
Number of aPower batteries	REQUIRED		☐

What it's for: Sizes every charge and discharge plan. Detected from the aGate; you only override it if detection is wrong.
Where to find it: FranklinWH app device list, or your installation paperwork.

Battery model REQUIRED ☐ aPower (original) ☐ aPower 2 ☐ Not sure

What it's for: Sets usable capacity per unit (aPower 2 is 13.6 kWh each). Total capacity is detected; this lets you check it.
Where to find it: Unit label, installer paperwork, or the FranklinWH app device details.

Backup reserve set in the FranklinWH app (%) REQUIRED

What it's for: The charge kept for outages. The Franklin app is the source of truth; the engine plans never to discharge below it.
Where to find it: FranklinWH app, in the operating mode / reserve settings.

Hold a fixed reserve no matter what the app says? OPTIONAL

What it's for: Leave blank to let the app decide. Only set a value if several people change the app, or a utility / VPP program resets it; the automation then sends this value on every mode switch and overrides the app.
Where to find it: Your choice. Write the percentage, or leave blank.

3. Solar arrays - array 1

Fill one block per array. The key question is whether an array is wired into the aGate (it can charge the battery) or on its own meter (it only offsets the bill). If you have no solar, skip this section.

Item	Need	Your information	Matches review
Array name **What it's for:** A short name you will recognize on the dashboard, such as House or Barn. **Where to find it:** Your choice.	REQUIRED		☐
Brand / monitoring type **What it's for:** Decides how production is read: Enphase from the local gateway, SolarEdge from the monitoring portal. **Where to find it:** Inverter or gateway labels, or your installer paperwork.	REQUIRED	☐ Enphase ☐ SolarEdge ☐ Other / none	☐
Charges the battery? **What it's for:** Only arrays connected to the aGate solar input charge the battery. These drive the forecast and the engine. An array on a separate meter never charges the battery. **Where to find it:** Your installer or single-line diagram: is the array connected to the aGate, or to its own meter/panel?	REQUIRED	☐ Yes ☐ No ☐ Not sure	☐
Exports surplus to the grid? **What it's for:** Whether this array can send surplus power to the grid (some battery-charging arrays are non-export by design). **Where to find it:** Interconnection agreement or installer.	REQUIRED	☐ Yes ☐ No ☐ Not sure	☐
AC size (kW) **What it's for:** Inverter output limit. Used for sanity checks and the dashboard. **Where to find it:** Sum of inverter or microinverter AC ratings, from the permit set, the monitoring app, or installer paperwork.	REQUIRED		☐
DC size (kWp) Needed for a battery-charging array. **What it's for:** Panel nameplate total (panel count x panel watts). This is the solar forecast input. **Where to find it:** Permit plans or installer proposal: number of panels times the panel wattage.	IF APPLIES		☐
Tilt (degrees) Battery-charging array. **What it's for:** Panel angle from flat. Shapes the forecast curve. **Where to find it:** Permit plans, or estimate from roof pitch (a 4/12 roof is about 18 degrees, 6/12 about 27).	IF APPLIES		☐
Azimuth (degrees, 0 = south) Battery-charging array. **What it's for:** Direction the panels face: 0 south, -90 east, 90 west. A wrong azimuth skews the forecast toward morning or afternoon. **Where to find it:** Permit plans, or a compass / satellite map of the roof face.	IF APPLIES		☐
Enphase: gateway (Envoy / IQ Gateway) IP Enphase arrays. **What it's for:** Production is read locally from the Enphase gateway. **Where to find it:** Router's device list (often shows as Envoy). Reserve the address in the router.	IF APPLIES		☐
Enphase: gateway serial number Enphase arrays. **What it's for:** Needed to request the local access token. **Where to find it:** Label on the gateway, or the Enphase app / Enlighten device list.	IF APPLIES		☐
Enphase: Enlighten login email Enphase arrays. **What it's for:** Used once in a while to fetch the local gateway token. Stays in .env. **Where to find it:** The account you use for the Enphase app / Enlighten.	IF APPLIES		☐

Enphase: Enlighten password IF APPLIES
Enphase arrays.

☐ I have this (safer: tick this instead of writing it down)

What it's for: Used with the email above. Stays in .env.
Where to find it: Your Enlighten password.

Enphase: microinverter model OPTIONAL

What it's for: Shown on the dashboard and used for per-panel health scaling (for example IQ8MC).
Where to find it: Enphase app device list or the microinverter labels.

SolarEdge: site ID IF APPLIES
SolarEdge arrays.

What it's for: Identifies your system in the SolarEdge monitoring portal.
Where to find it: SolarEdge monitoring portal: the number in the site details or the site URL.

SolarEdge: API key IF APPLIES
SolarEdge arrays.

☐ I have this (safer: tick this instead of writing it down)

What it's for: Lets the automation read production from the portal. Stays in .env.
Where to find it: Monitoring portal, site Admin, then Site Access, API access (site admin rights needed; if your installer owns the site, ask them).

3. Solar arrays - array 2

Item	Need	Your information	Matches review

Array name REQUIRED

What it's for: A short name you will recognize on the dashboard, such as House or Barn.
Where to find it: Your choice.

Brand / monitoring type REQUIRED ☐ Enphase ☐ SolarEdge ☐ Other / none

What it's for: Decides how production is read: Enphase from the local gateway, SolarEdge from the monitoring portal.
Where to find it: Inverter or gateway labels, or your installer paperwork.

Charges the battery? REQUIRED ☐ Yes ☐ No ☐ Not sure

What it's for: Only arrays connected to the aGate solar input charge the battery. These drive the forecast and the engine. An array on a separate meter never charges the battery.
Where to find it: Your installer or single-line diagram: is the array connected to the aGate, or to its own meter/panel?

Exports surplus to the grid? REQUIRED ☐ Yes ☐ No ☐ Not sure

What it's for: Whether this array can send surplus power to the grid (some battery-charging arrays are non-export by design).
Where to find it: Interconnection agreement or installer.

AC size (kW) REQUIRED

What it's for: Inverter output limit. Used for sanity checks and the dashboard.
Where to find it: Sum of inverter or microinverter AC ratings, from the permit set, the monitoring app, or installer paperwork.

DC size (kWp) IF APPLIES
Needed for a battery-charging array.

What it's for: Panel nameplate total (panel count x panel watts). This is the solar forecast input.
Where to find it: Permit plans or installer proposal: number of panels times the panel wattage.

Tilt (degrees) IF APPLIES
Battery-charging array.

What it's for: Panel angle from flat. Shapes the forecast curve.
Where to find it: Permit plans, or estimate from roof pitch (a 4/12 roof is about 18 degrees, 6/12 about 27).

Azimuth (degrees, 0 = south) — IF APPLIES
Battery-charging array.
What it's for: Direction the panels face: 0 south, -90 east, 90 west. A wrong azimuth skews the forecast toward morning or afternoon.
Where to find it: Permit plans, or a compass / satellite map of the roof face.

Enphase: gateway (Envoy / IQ Gateway) IP — IF APPLIES
Enphase arrays.
What it's for: Production is read locally from the Enphase gateway.
Where to find it: Router's device list (often shows as Envoy). Reserve the address in the router.

Enphase: gateway serial number — IF APPLIES
Enphase arrays.
What it's for: Needed to request the local access token.
Where to find it: Label on the gateway, or the Enphase app / Enlighten device list.

Enphase: Enlighten login email — IF APPLIES
Enphase arrays.
What it's for: Used once in a while to fetch the local gateway token. Stays in .env.
Where to find it: The account you use for the Enphase app / Enlighten.

Enphase: Enlighten password — IF APPLIES
Enphase arrays.
☐ I have this (safer: tick this instead of writing it down)
What it's for: Used with the email above. Stays in .env.
Where to find it: Your Enlighten password.

Enphase: microinverter model — OPTIONAL
What it's for: Shown on the dashboard and used for per-panel health scaling (for example IQ8MC).
Where to find it: Enphase app device list or the microinverter labels.

SolarEdge: site ID — IF APPLIES
SolarEdge arrays.
What it's for: Identifies your system in the SolarEdge monitoring portal.
Where to find it: SolarEdge monitoring portal: the number in the site details or the site URL.

SolarEdge: API key — IF APPLIES
SolarEdge arrays.
☐ I have this (safer: tick this instead of writing it down)
What it's for: Lets the automation read production from the portal. Stays in .env.
Where to find it: Monitoring portal, site Admin, then Site Access, API access (site admin rights needed; if your installer owns the site, ask them).

4. Rates

The rate plan on the meter the battery is behind. Peak windows for the engine come from here. Most time-of-use plans have summer and winter seasons with different prices, and sometimes different windows.

Item	Need	Your information	Matches review
Utility and rate plan name	REQUIRED		

What it's for: Which plan the engine optimizes for (for example PG&E EV2-A).
Where to find it: Your bill, usually labelled "Rate schedule" on the electric charges page.

Prices and windows by season REQUIRED Use the grid below. ☐

Season (months)	Tier	Price (cents/kWh)	Window(s), e.g. 16:00-21:00 weekdays
	Peak		
	Partial-peak		
	Off-peak		
	Peak		
	Partial-peak		
	Off-peak		

What it's for: Price per kWh for each tier, and the clock times each tier applies. The engine charges before peak and covers the peak window from the battery.
Where to find it: Your utility's published tariff sheet for the plan (search the plan name plus "tariff"). Use total delivered price per kWh if your bill splits generation and delivery. Note weekday-only windows.

Holidays treated as off-peak? OPTIONAL ☐ Yes ☐ No ☐ Not sure ☐

What it's for: Some plans price listed holidays as off-peak all day.
Where to find it: Tariff sheet, usually a footnote under the time periods.

Where to find your official rate schedule

Your bill names the rate schedule (for example EV2-A or E-TOU-D). The utility's tariff document for that schedule is the authority for prices, windows and seasons; compare the review against it. PG&E is listed below; links for other utilities will be added as we find them.
PG&E (California): https://www.pge.com/tariffs/en.html#ELECTRIC%20RATE%20SCHEDULES
Open Electric Rate Schedules. Each residential schedule is a PDF (for example EV2 or E-TOU-D); the price table and the time periods are on the first pages.

5. Utility & billing

Item	Need	Your information	Matches review

Discount program on this meter REQUIRED ☐ None ☐ CARE ☐ FERA ☐ Other ☐

What it's for: Income-qualified discounts change the price per kWh. The discounted prices go in the rate plan; this records which program applies.
Where to find it: Your bill lists the program if you are enrolled. Programs end; check it is still active.

How exports are credited REQUIRED ☐ Net metering (1:1 retail) ☐ Net billing (NEM3-style) ☐ No export ☐

What it's for: Tells the engine whether exported energy is worth close to retail or much less. In California, NEM 1 and NEM 2 are net metering; NEM 3 / NBT is net billing.
Where to find it: Your interconnection approval letter, or the NEM line on your bill.

Second meter billed together with this one? OPTIONAL ☐ Yes ☐ No ☐ Not sure ☐

What it's for: For aggregate billing, such as a separately metered array whose exports offset this account.
Where to find it: Your bill shows multiple service IDs / meters under one account.

6. Engine behavior

Choices rather than facts. The defaults suit most installs; note what you would like.

Item	Need	Your information	Matches review

Strategy for surplus solar REQUIRED ☐ Self-consumption (store it) ☐ Export (sell it) ☐

What it's for: Self-consumption keeps surplus solar in the battery and caps grid charging so solar can finish the fill. Export skips that and lets surplus go to the grid for credit. Non-export systems and net billing usually want self-consumption.
Where to find it: Your decision. Match it to your export credit (section 5).

Highest charge from solar (%) — OPTIONAL

Default 95

What it's for: How full solar may fill the battery.
Where to find it: Leave at the default unless you have a reason.

Highest charge from the grid (%) — OPTIONAL

Default 85

What it's for: Grid charging stops here. Above it the charge rate tapers, so solar is left to finish. Lower it if you see solar being curtailed on full-battery days.
Where to find it: Leave at the default to start.

7. Forecast, weather & site

Item	Need	Your information	Matches review
Site latitude, longitude	REQUIRED		

What it's for: Drives the solar forecast (free Open-Meteo service, no key needed). Two decimal places is plenty.
Where to find it: Right-click your home in Google Maps; the first line is latitude, longitude.

Weather Underground station ID — OPTIONAL

What it's for: A personal weather station (PWS) near you improves forecast calibration. It can be your own station or any nearby public one; the review test shows how far it is from your site.
Where to find it: wunderground.com/wundermap: pick a station close to you, for example KCAGEORG2. Without a station and key, leave this off.

Weather Underground API key — IF APPLIES
Needed if you use a station.

I have this (safer: tick this instead of writing it down)

What it's for: Lets the automation read the station. Stays in .env.
Where to find it: wunderground.com/member/api-keys (free keys are tied to owning a station).

8. Integrations (all optional)

Item	Need	Your information	Matches review
PVOutput API key	OPTIONAL		

I have this (safer: tick this instead of writing it down)

What it's for: Uploads daily production to PVOutput.org. Stays in .env.
Where to find it: pvoutput.org/account.jsp (enable API access there).

PVOutput system ID(s) — IF APPLIES
If you use PVOutput.

What it's for: One ID per array, in the same order as your arrays.
Where to find it: pvoutput.org, your systems list (the "sid" number).

SolarEdge portal login (panel health) — OPTIONAL

What it's for: Per-optimizer health and degradation tracking from the SolarEdge portal. This is the portal login, not the API key.
Where to find it: The email you use for monitoring.solaredge.com.

SolarEdge portal password — IF APPLIES
If you use panel health.

I have this (safer: tick this instead of writing it down)

What it's for: Stays in .env.
Where to find it: Your SolarEdge portal password.

SolarEdge inverters: local Modbus IPs **What it's for:** Reads SolarEdge inverters directly on your network (SunSpec Modbus TCP, usually port 1502). Only if Modbus TCP is enabled on the inverters. **Where to find it:** Enabled with SolarEdge SetApp or by your installer; IPs from the router device list.	OPTIONAL		☐
Real-time (hourly) pricing provider **What it's for:** Only for hourly-priced plans such as ComEd. Most time-of-use customers leave this off. **Where to find it:** Your utility plan documents.	OPTIONAL		☐

9. Telemetry

Item	Need	Your information	Matches review
Share anonymous usage telemetry? **What it's for:** A daily anonymous summary that helps the project see which features and configurations are in use. No credentials, IPs, serials or exact location. **Where to find it:** Your choice.	OPTIONAL	☐ Yes ☐ No ☐ Not sure	☐

Advanced settings

Advanced settings (polling and decision intervals, retries, timeouts, debug logging, the v3 legacy options) have safe defaults and are not on this list. The review shows them under Advanced if you want to look.

Notes (testers: anything missing, unclear, or different in the review)